

Laptop Battery Health Analysis

https://gemini.google.com/app/a4a3824902be0553

User prompt: is this good or bad battery present: yes rechargeable: yes state: discharging warning-level: none energy: 39.45 Wh energy-empty: 0 Wh energy-full: 47.78 Wh energy-full-design: 47 Wh energy-rate: 7.868 W voltage: 12.124 V charge-cycles: 25 time to empty: 5.0 hours percentage: 83% capacity: 100% technology: lithium-polymer

Response: This output shows a **very healthy battery in optimal condition**.

Core Health Metrics

- **Capacity (100%):** Your battery's current full charge capacity (47.78 Wh) exceeds its original design capacity (47 Wh). This means zero degradation—your battery is performing at 100% of its factory specification.
- **Charge Cycles (25):** Very low usage. Most modern lithium-polymer batteries are rated for 300 to 500 charge cycles before experiencing noticeable degradation.
- **State & Power Draw:**
 - **Energy Rate (7.87 W):** Low power consumption for active use, indicating good system power efficiency.
 - **Time to Empty (5.0 hours at 83%):** Translates to ~6 hours of total run time on a full charge, which is strong performance.

Key Data Breakdown

Metric	Value	Status
Full Charge / Design	47.78 Wh / 47.00 Wh	Excellent (No wear)
Health Capacity	100%	Optimal
Cycle Count	25	Very Low
Discharge Rate	7.87 W	Efficient

User prompt: How can I maintain my laptop battery health long-term on Linux?

Response: Maintaining long-term battery health on Linux comes down to two main goals: **limiting heat generation** and **preventing high-voltage stress** caused by keeping the battery at 100% state of charge continuously.

1. Set Charge Thresholds (Most Critical)

Lithium-ion and lithium-polymer batteries degrade fastest when held at 100% charge or left plugged in at full capacity. Limiting maximum charge to **80%** significantly extends lifespan.

- **For Lenovo Laptops (via TLP or Sysfs):** Check if charge thresholds are supported on your system by running:

```
sudo tlp-stat -b
```


To set charging to stop at 80% and resume at 75%:

```
sudo tlp setcharge 75 80
```

(Note: Hardware support varies by model family; ThinkPads have full kernel support, while some IdeaPad/V-series models use `ideapad_laptop` kernel modules for Conservation Mode).

- **Enable Conservation Mode directly (if supported):**

```
echo 1 | sudo tee /sys/bus/platform/drivers/ideapad_acpi/vocal_mode
# Or via ideapad_laptop driver:
echo 1 | sudo tee /sys/bus/platform/drivers/ideapad_laptop/conservation_mode
```

Verify success: If the value returns 1, your battery will cap charging around 60–80%.

2. Power Optimization & Thermal Control

Heat is the primary accelerator of chemical degradation inside battery cells.

- **Install and Enable TLP:** TLP automatically optimizes system settings (PCIe power saving, CPU scaling, disk spin-down) when running on battery power.

```
sudo apt install tlp tlp-rdw
sudo systemctl enable --now tlp
```

- **Use Auto-CPUFreq for Dynamic Governor Switching:** Combines power efficiency when idle with performance under load without manual governor tweaks:

```
sudo apt install auto-cpufreq
sudo auto-cpufreq --install
```

- **Disable Turbo Boost on Battery (Optional):** Prevent brief power spikes and sudden temperature jumps during light tasks:

```
# For Intel CPUs
echo 1 | sudo tee /sys/devices/system/cpu/intel_pstate/no_turbo
```

3. Best Operational Practices

Practice	Recommendation	Why
Ideal Charge Window	Keep between 20% and 80%	Reduces voltage stress on internal cell chemistry.
Avoid Deep Discharges	Charge before hitting 15%	Complete discharges (0%) place severe stress on modern lithium cells.
Manage Heavy Loads	Plug in during compilation / heavy rendering	High current draw on battery causes rapid heating.
Storage State	Store around 50% charge	If unused for weeks, ~50% charge minimizes capacity loss over time.

User prompt: `sudo: tlp-stat: command not found`

Response: The `tlp-stat` command is missing because the **TLP package is not installed** on your system yet.

Here is how to install TLP, configure it, and verify that it works.

1 Install TLP

Prerequisite

Update your package repository list and install `tlp` along with its radio device management wizard (`tlp-rdw`):

```
sudo apt update
```

```
sudo apt install tlp tlp-rdw -y
```

2 Enable and Start the TLP Service

Service Management

Enable TLP so it starts automatically on boot, then start the service immediately:

```
sudo systemctl enable --now tlp
```

Verification: Run `systemctl status tlp` and look for active (running).

3 Check Battery Statistics & Threshold Support

Diagnostics

Now that TLP is installed, run the battery statistics command again:

```
sudo tlp-stat -b
```

Verification: Look under the `+++ Battery Care` or `+++ Battery Status` section in the output. It will explicitly show your current charge thresholds and whether setting thresholds is supported on your hardware.

```
User prompt: moss@anhedonia:~$ sudo tlp-stat -b --- TLP 1.6.1 ----- +++ Battery Care Plugin:
lenovo Supported features: charge threshold Driver usage: * vendor (ideapad_laptop) = active (charge threshold)
Parameter value range: * STOP_CHARGE_THRESH_BAT0: 0(off), 1(on) -- battery conservation mode /sys/bus/platform/
drivers/ideapad_acpi/VPC2004:00/conservation_mode = 0 (100%) +++ Battery Status: BAT0 /sys/class/power_supply/BAT0/
manufacturer          = SMP /sys/class/power_supply/BAT0/model_name          = L22M3PF2 /sys/class/
power_supply/BAT0/cycle_count      = 25 /sys/class/power_supply/BAT0/energy_full_design    = 47000
[mWh] /sys/class/power_supply/BAT0/energy_full      = 47780 [mWh] /sys/class/power_supply/BAT0/energy_now
          = 36150 [mWh] /sys/class/power_supply/BAT0/power_now          = 11053 [mW] /sys/class/power_supply/
BAT0/status              = Discharging Charge              = 75.7 [%] Capacity
          = 101.7 [%]
```

Response: Your output confirms that **battery conservation mode is fully supported** on your system via the `ideapad_laptop` driver, but it is currently turned off (`conservation_mode = 0`).

On Lenovo laptops using this driver, conservation mode caps charging at **60%** (or ~80% depending on firmware revision) to preserve battery life when plugged in.

1 Enable Conservation Mode via TLP

Recommended

Run the TLP command to turn on battery conservation mode:

```
sudo tlp setcharge 1
```

Verification: Run `sudo tlp-stat -b | grep conservation_mode` to confirm it now shows `= 1`.

2 Make Conservation Mode Persistent Across Reboots

Configuration

To ensure this setting remains active every time you boot:

1. Open the TLP configuration file in your text editor:

```
sudo nano /etc/tlp.conf
```

2. Search for `STOP_CHARGE_THRESH_BAT0` (press `Ctrl + W` in nano).

3. Uncomment the line (remove the `#`) and set it to 1:

```
STOP_CHARGE_THRESH_BAT0=1
```

4. Save and exit (Ctrl + O, Enter, then Ctrl + X).

3 Alternative Direct Kernel Switch

Instant Toggle

If you ever want to toggle it manually without TLP:

- **Enable conservation mode:**

```
echo 1 | sudo tee /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode
```

- **Disable (allow 100% full charge for travel):**

```
echo 0 | sudo tee /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode
```

Verification: Plug in your AC adapter when battery is above 60–80%. The battery status should read **"Plugged in, not charging"** or **"Idle"**.

User prompt: moss@anhedonia:~\$ sudo tlp setcharge 1 Error: battery 1 not present.

Response: The `tlp setcharge` command expects a threshold percentage (like 75–80) or explicit battery identifiers on standard ThinkPad drivers. On Lenovo IdeaPad/V-series hardware using the `ideapad_laptop` driver, the parameter name is `STOP_CHARGE_THRESH_BAT0=1`, but the CLI command syntax differs.

1 Enable Conservation Mode directly via sysfs

Instant fix

Apply the setting directly to the kernel sysfs interface path revealed in your `tlp-stat` output:

```
echo 1 | sudo tee /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode
```

Verification: Run `cat /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode`. It should return 1.

2 Configure TLP for persistence across reboots

System configuration

Open the TLP config file to ensure this setting applies automatically on every system boot:

```
sudo nano /etc/tlp.conf
```

Find `STOP_CHARGE_THRESH_BAT0` (press Ctrl + W to search), uncomment it (remove #), and set it to 1:

```
STOP_CHARGE_THRESH_BAT0=1
```

Save and exit (Ctrl + O, Enter, then Ctrl + X).

3 Apply TLP configuration

Service restart

Reload TLP to enforce the configuration file:

```
sudo tlp start
```

Verification: Plug in your charger while battery is above 60–80%. The battery status in your tray or `upower -i /org/freedesktop/UPower/devices/battery_BAT0` will display **"Pending charge"**, **"Not charging"**, or **"Idle"**.

User prompt: `STOP_CHARGE_THRESH_BAT0=80`

Response: For your specific Lenovo laptop hardware (which uses the `ideapad_laptop` / `ideapad_acpi` vendor driver), the valid parameter value range is **binary: 0 or 1**, not a raw percentage like 80.

Setting `STOP_CHARGE_THRESH_BAT0=80` in TLP will cause an invalid parameter error on this driver.

Correct Configuration

1. Open `/etc/tlp.conf`:

```
sudo nano /etc/tlp.conf
```

2. Set the threshold line to **1**:

```
STOP_CHARGE_THRESH_BAT0=1
```

3. Save, exit (Ctrl + O, Enter, then Ctrl + X), and restart TLP:

```
sudo tlp start
```

What value 1 actually does

When set to 1, your laptop firmware engages **Conservation Mode**. Depending on your Lenovo model's embedded controller, this automatically caps charging at either **60% or 80%** (most modern Lenovo V-series/IdeaPad models stop around 75–80%).

To verify if charging is successfully capped:

```
cat /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode
```

If it outputs 1, Conservation Mode is active.

User prompt: *it was `STOP_CHARGE_THRESH_BAT0=80` by def*

Response: You are completely right to point that out—in `/etc/tlp.conf`, the default template text for ThinkPads and general laptops is indeed commented out as `#STOP_CHARGE_THRESH_BAT0=80`.

The difference here comes down to **how TLP translates settings depending on your specific laptop driver**:

How TLP Handles This Driver

On standard Lenovo ThinkPads (using the `tp_smapi` or `thinkpad_acpi` drivers), TLP accepts arbitrary percentage values like 80.

However, on Lenovo V-series, IdeaPad, and Legion models (which use the `ideapad_laptop` driver exposed at `/sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode`), the hardware firmware **only accepts an ON/OFF toggle (1 or 0)**.

TLP 1.6+ is smart enough to handle this mapping:

- When you set `STOP_CHARGE_THRESH_BAT0=1` (or any non-zero value like 80), TLP writes 1 to the `conservation_mode` kernel node.
- Your laptop's embedded controller then handles capping the charge (usually around 75–80% or 60% depending on firmware).

Quick Verification

To confirm that changing `/etc/tlp.conf` worked and activated the hardware cap:

```
sudo tlp start
cat /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode
```

- **Expected Output:** 1
- **How to verify success:** Plug in your AC adapter while your battery is above 75–80%. Run `sudo tlp-stat -b`. Under `+++ Battery Status: BAT0`, the status line will change from `Charging` to **Not charging** or **Idle**.

User prompt: so my laptop now will stop charging at 80 and continue at 70, is that 100% healthier and extend battery life?

Response: Yes, absolutely! Capping your battery charge significantly extends its overall lifespan.

While no battery lasts forever, keeping lithium-polymer cells away from maximum voltage stress changes their wear rate dramatically.

Why Capping at 80% Works

- 1. Eliminates High-Voltage Stress:** Lithium cells experience the most chemical and structural degradation when held above **4.10 V per cell** (which corresponds to roughly 85%–100% state of charge). Capping at 80% keeps the voltage in a stable "sweet spot" (~3.8 V – 4.0 V).
- 2. Drastically Increases Cycle Life:**
 - **0% to 100% cycles:** A standard lithium-polymer battery typically lasts **300 to 500 full cycles** before losing 20% of its original capacity.
 - **20% to 80% cycles:** Operating within this partial range can boost total usable lifetime to **1,500–2,500+ charge cycles** before experiencing equivalent degradation.
- 3. Prevents Heat Degradation While Plugged In:** When a laptop stays plugged into AC power at 100%, high continuous voltage combined with system heat accelerates chemical degradation. Capping at 80% allows the power supply to run the laptop directly while letting the battery sit idle safely.

Summary Comparison

Strategy	Typical Lifespan	Degradation Speed
Always Charged to 100% (AC constantly)	~1.5 to 2 years	High (capacity drops quickly)
Capped at ~80% (Conservation Mode)	4 to 6+ years	Minimal

Is there any downside?

The only trade-off is **20% less runtime when unplugged**. Whenever you know you'll be traveling or working away from an outlet all day, you can simply disable conservation mode temporarily (`echo 0 | sudo tee /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode`) to top it up to 100%.

User prompt: i want that to stop and to return back charging to 100%

Response: To disable Conservation Mode and allow your battery to charge back up to 100%, you just need to set the value back to **0**.

- 1 Disable Conservation Mode via Sysfs**

Instant Change

Run this command in your terminal to allow charging up to 100% immediately:

```
echo 0 | sudo tee /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode
```

Verification: Check the current mode by running:

```
cat /sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode
```

It should return 0.

2 Update TLP Configuration

Persistence

To keep it charging to 100% permanently (even after restarting your laptop):

1. Open the TLP configuration file:

```
sudo nano /etc/tlp.conf
```

2. Change `STOP_CHARGE_THRESH_BAT0=1` back to 0 (or comment it out by adding # at the beginning):

```
STOP_CHARGE_THRESH_BAT0=0
```

3. Save and exit (Ctrl + O, Enter, then Ctrl + X).

4. Restart TLP:

```
sudo tlp start
```

Verification: Plug in your AC adapter. Your battery will now charge past 80% all the way to **100%**.

User prompt: How can I create a simple bash script to easily switch between 80% conservation mode and 100% full charge?

Response: Here is a lightweight bash script and terminal command alias to toggle Lenovo Conservation Mode on and off.

1 Create the Toggle Script

1 min

Create a new script file named `battery-toggle` in `/usr/local/bin`:

```
sudo nano /usr/local/bin/battery-toggle
```

Paste the following code inside:

```
#!/bin/bash
NODE="/sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode"

if [ ! -f "$NODE" ]; then
    echo "Error: Conservation mode sysfs node not found."
    exit 1
fi

CURRENT=$(cat "$NODE")

if [ "$CURRENT" -eq 0 ]; then
    echo 1 | sudo tee "$NODE" > /dev/null
    echo "Conservation Mode ENABLED (Capped at ~80%)."
```

```
else
    echo 0 | sudo tee "$NODE" > /dev/null
    echo "Conservation Mode DISABLED (Full charge to 100%)."
```

```
fi
```

Save and exit (Ctrl + O, Enter, then Ctrl + X).

2 Make the Script Executable

Permissions

Grant execution permissions to the script:

```
sudo chmod +x /usr/local/bin/battery-toggle
```

Verification: Run `battery-toggle` in your terminal. It will toggle the state and print whether Conservation Mode is enabled or disabled.

3 Add a Quick Shell Alias

Optional

If you want an even shorter command without typing `battery-toggle`, add an alias to your `~/.bashrc` file:

```
echo "alias batt='battery-toggle'" >> ~/.bashrc
source ~/.bashrc
```

Verification: Type `batt` in your terminal to switch between 80% and 100% modes.

User prompt: Does that script include `STOP_CHARGE_THRESH_BAT0` i commented out?

Response: No, that script **does not use or modify** `STOP_CHARGE_THRESH_BAT0` in your TLP configuration file.

Instead, the script bypasses TLP entirely and writes directly to the Linux kernel interface: `/sys/bus/platform/drivers/ideapad_acpi/VPC2004:00/conservation_mode`

How they interact:

- **Your TLP Config (`/etc/tlp.conf`):** Since `STOP_CHARGE_THRESH_BAT0` is commented out, TLP will not enforce any charge limit on system boot or when restarting TLP.
- **The `battery-toggle` Script:** Toggles the hardware setting directly on the fly whenever you run it.

The change made by running the script takes effect immediately in real time, regardless of what is commented out inside your `/etc/tlp.conf` file.

User prompt: is `tlp.conf` important

Response: Yes, **`tlp.conf` is important** if you want TLP to automatically manage your power settings, performance profiles, and battery rules on Linux without manual intervention.

Here is what makes `tlp.conf` key to your system:

Key Roles of `tlp.conf`

1. Automation Across Boots & Power States

- While manual commands or `sysfs` writes (like your `battery-toggle` script) change hardware settings temporarily, **`/etc/tlp.conf` is what TLP reads during boot and every time you switch between AC power and battery power.**
- It ensures your preferred power management policies are enforced automatically.

2. Centralized Hardware & Power Controls

Beyond battery charge limits, `tlp.conf` controls nearly every power-saving aspect of your laptop:

- **CPU Performance:** Sets CPU frequency governors and energy-performance preferences separately for AC vs. Battery.
- **PCIe & GPU Power Saving:** Controls power states for discrete graphics cards and internal components.
- **Drive & USB Power Management:** Automatically suspends idle USB devices and sets hard drive / NVMe power states.
- **Wireless Devices:** Enables or disables Wi-Fi, Bluetooth, and WWAN depending on power state or dock status.

Do you need it if you use the battery-toggle script?

- **For Battery Conservation Mode:** No. Because you commented out `STOP_CHARGE_THRESH_BAT0` and use the direct `battery-toggle` script, `tlp.conf` won't override your conservation setting.
 - **For Overall Laptop Power Savings:** Yes. TLP still reads `tlp.conf` to optimize CPU behavior, device power saving, and overall power efficiency in the background whenever you unplug your charger.
-